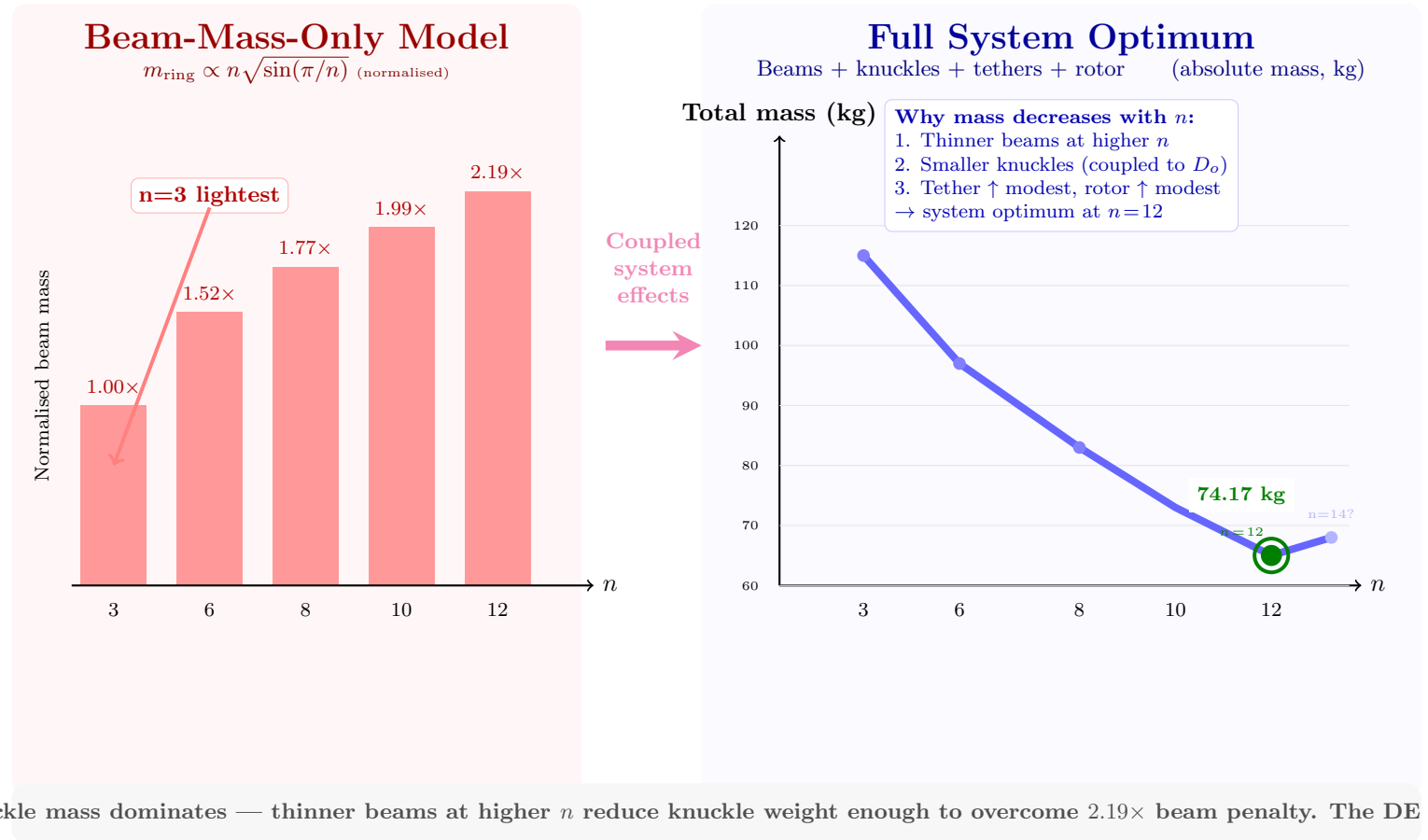


Why the Optimiser Chose $n = 12$ Despite Beam-Only Scaling

Beam mass alone says fewer lines are lighter. Coupled system effects reverse the tradeoff.



Counterintuitive result: Beam-only (left) says $n = 3$ lightest. But coupled knuckle mass dominates — thinner beams at higher n reduce knuckle weight enough to overcome 2.19× beam penalty. The DE optimiser searched